

Cumulant Based Stochastic Reactive Power Planning Method for Distribution Systems With Wind Generators

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Abstract—Proliferation of wind generators (WGs) requires a change in distribution system planning techniques as WGs have intermittent and uncertain output. This paper proposes a Cumulant based stochastic optimal reactive power planning method for distribution systems with high penetration of WGs. Uncertainties in the output of WGs and load forecasts are modeled using probability density functions (PDFs). With a stochastic framework, an optimization challenge is formulated to minimize the total costs of new capacitors and the total annual energy loss. The optimization problem is then solved by using the Logarithmic Barrier Interior Point Method (LBIPM). At the optimal solution, LBIPM provides a linear relationship between the cumulants of independent variables (load and wind power) and the cumulants of the dependent system parameters. The Cumulant method offers a generous advantage in speed, while maintaining acceptable accuracy, as compared to the computationally cumbersome traditional Monte Carlo simulation (MCS) method. The method is tested on 7-bus, 33-bus, and 129-bus systems. The results are reported and discussed. The performance and accuracy are assessed by comparing the results with those from MCS method.

Index Terms—Capacitor planning, Cumulant method, distribution system planning, stochastic programming.

NOMENCLATURE

$f_x(x)$	PDF of random variable.
$\overline{PD_i}$	Mean of the load demand (MW).
σ^2	Variance of random variable.
ρ_{w_i}	PDF of wind speed.
γ	Shape factor of Weibull distribution.
β	Scale factor of Weibull distribution.
w_i	Wind speed in m/s.
KC	Annual cost of per unit capacitor bank installed in an RDS (\$/kVAR).
QS^d	Bus-wise vector of new capacitors at dth time interval.

KL	Annual cost of per unit energy loss (\$/kWh).
TD_d	Time interval for dth interval.
LF^d	Load factor at dth interval.
PD, QD	Vector of bus-wise real and reactive power loads.
PW	Vector of real power output of WGs scale.
PL	System real power loss at dth load level.
SL	Vector of line MVA flows at dth load level.
V^d	Vector of bus voltages magnitudes at dth load level.
P^d, Q^d	Vector of receiving end power in lines at dth load level.
FV, FP, FQ	3NB equations (voltage and receiving end power) representing a distribution system.
NB	Number of buses in the system.
ND	Number of segments in load duration curve.
NC	Number of candidate buses.
PB	Difference between generation and load.
$L(u)$	Lagrangian function.
U	Vector of primal and dual variables in the optimization problem.
λ, μ	Lagrangian multiplier.
J_g	Jacobian of the equality constraints, $g(x)$.
J_h	Jacobian of the inequality constraints, $h(x)$.
$\nabla f(X)$	Gradient of the objective function, $f(x)$.
H	Hessian of the Lagrangian.
$\nabla L(U)$	Gradient of the Lagrangian.
ΔPB	Vector of changes in bus generation and loading.
Z	Input random variable.
Y	Output random variable.
C_m	mth coefficient in the linear combination.
s	Complex frequency.
$\Phi(s)$	Moment generating function.

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$\Psi(s)$	Cumulant generating function.
k_r	r th cumulants of random variable.
PDF	Probability density function.
RDS	Radial distribution systems.
WG	Wind generator.
MCS	Monte Carlo simulations.
LBIPM	Logarithmic-Barrier Interior Point Method.
KKT	Karush-Kuhn-Tucker conditions.

I. INTRODUCTION

SHUNT capacitor banks are used in distribution systems to improve voltage profile and reduce energy losses. Optimally sizing and allocating capacitor banks has been a common task in distribution system planning for many decades. Several optimization methods have been proposed to solve this problem. Analytical methods were the first to gain popularity [1]–[3], followed by iterative methods [4]–[7], and in recent years there has been the emergence of artificial intelligence-based methods [8]–[12]. Nowadays, however, the typical distribution system configuration has been changing due to the large-scale integration of wind generators (WGs). Capacitor planning in these modern distribution systems has not been well explored considering all uncertainties inherent to WGs. The uncertainties related to the wind availability and load consumption can be addressed by a probabilistic approach which takes into account these uncertainties by using probabilistic density functions (PDFs).

To deal with such stochastic problems, Monte Carlo simulation (MCS) technique can be used in order to cover all possible scenarios of load demand and WGs availability so that an optimal capacitor plan can be determined. However, the nature of these variables would result in a very large scale combinatorial investigation, rendering the solution computationally inefficient due to the exhaustive exploratory nature of the MCSs. The situation would greatly worsen for larger systems with high penetration of WGs. In order to consider all uncertainties, the authors of [13] formulated a Fuzzy stochastic capacitor planning method to minimize the total cost of capacitor banks as well as the annual energy loss in a distribution system by using probabilistic models of load and WGs as Gaussian (Normal) and Weibull PDFs, respectively. Then the proposed fuzzy stochastic programming method was linearized to be successively solved using a robust Mixed Integer Linear Programming Solver to find the most probable optimal value.

However, it does not provide any probabilistic information of the optimized output variables, and moreover, as the number of generated samples increases, the execution time for finding the most optimal values increases as well. That can be very time consuming for large systems.

We propose in this paper, as an alternative to the idea presented in [13], a probabilistic optimal capacitor planning method using Cumulant technique which gives probabilistic information of optimized variables such as voltages and capacitor sizes. As in [13], load forecasts are modeled as Gaussian

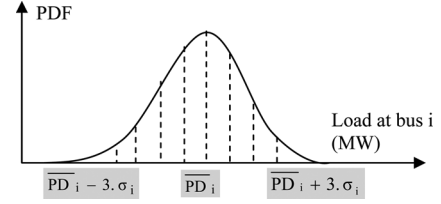


Fig. 1. PDF of the annual load demand (Gaussian distribution).

PDFs and WG outputs are modeled as Weibull PDFs. Once formulated, the optimization problem is solved by using the Logarithmic Barrier Interior Point Method (LBIPM), which provides a linear relationship between the cumulants of input variables (load demand and wind availability) and those of output variables (bus voltages, active power, reactive power, and capacitor size). Finally, the output variables have their PDFs reconstructed by using the Gram-Charlier Expansion theory. To assess efficacy of the proposed method, it is tested on three different size systems, i.e., small, medium and large IEEE test systems (7-bus, 33-bus, and 129-bus systems). The performance and accuracy are assessed by comparing these results with those from MCS technique with 5000 samples.

This paper is structured as follows. Section II presents the modeling of load demand and wind generation using PDFs. Section III presents the capacitor allocation problem formulation. Section IV introduces the solution method. Section V reports and discusses the results of the proposed method applied to the test systems. Section VI concludes the paper.

II. PROBABILISTIC MODELING OF LOAD AND WIND GENERATORS

Power drawn from a distribution system depends on consumer usage patterns, whereas the WG power production depends on the wind availability. Therefore, load demand and WG output are random quantities, and as such, can be represented by probabilistic models.

A. Probabilistic Modeling of Loads

The behavior of the load consumption has been modeled by many researchers. This paper follows the idea presented in [13], which modeled the annual load demand as a Normal distribution. The associated PDF is illustrated in Fig. 1 and its characteristics are given by

$$f_x(x) = \left[\frac{1}{\sqrt{2\pi\sigma_i^2}} \right] \times e^{-(PD_i - \overline{PD}_i)^2 / 2\sigma_i^2} \quad (1)$$

where $\overline{PD}_i$ is the mean and σ_i is the standard deviation of the load demand (MW) at bus i . For convenience, the probabilities associated with reactive power in this work are modeled as Gaussian distribution.

B. Probabilistic Modeling of Wind Generation

The probabilistic distribution of a wind speed has been researched and published by many authors in the last two decades. Following the idea from [14], in this paper we model the annual

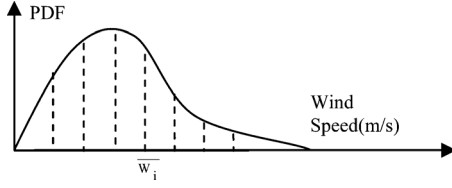


Fig. 2. PDF of wind speed (Weibull distribution).

wind speed PDF for the purposes of this planning exercise using a Weibull distribution such as that shown in Fig. 2.

Mathematically, it is written as

$$p_{w_i} = \frac{\gamma}{\beta_i^\gamma} w_i^{\gamma-1} \times e^{-(w_i/\beta_i)^\gamma} \quad (2)$$

where w_i is the wind speed in m/s, and γ and β_i are the shape and scale parameters of the Weibull distribution, respectively. If more than one WG is installed in the same geographical area, it is reasonable to assume they all have the same γ due to proximity.

In this work, power generated by wind turbines and other chronological aspects are not correlated and treated as independent.

III. PROBLEM FORMULATION

This section formulates the capacitor planning optimization problem for distribution systems. The goal of this research is to identify the optimal sizes and locations for capacitor banks such that the sum of the total cost of the newly sited and sized capacitors, as well as the annual energy losses, are minimized while maintaining the voltage profile within a prescribed limit. Once a capacitor unit is sited at a bus along the feeder, it changes the feeder nodal power and voltages, leading to a reduction in real power loss in the system. The complete non-linear problem can be defined as follows with problem vector $X^d = [QS^d, V^d, P^d, Q^d]$ and $X = [X^1, X^2, \dots, X^{ND}]$.

Objective Function:

Minimize cost of new capacitors and annual energy loss costs:

$$TC = KC \times \sum_{i=1}^{NC} \left(\max_d \{QS_i^d\} \right) + KL \times \sum_{d=1}^{ND} TD_d \times PL(V^d, P^d, Q^d). \quad (3)$$

Constraints:

(a) Bus Power Balance Equations [15], $\forall d = 1$ to ND

$$\text{Voltage Equation: } FV(V^d, P^d, Q^d) = 0 \quad (4)$$

$$\text{P-Power Balance: } FP(V^d, P^d, Q^d) = LF^d \cdot PD - PW \quad (5)$$

$$\text{Q-Power Balance: } FQ(V^d, P^d, Q^d) = LF^d \cdot QD - QS^d. \quad (6)$$

(b) Limits on problem variables

$$V_{\min} \leq V^d \leq V_{\max} \quad \forall d = 1 \text{ to } ND \quad (7)$$

$$QS_{\min} \leq QS^d \leq QS_{\max} \quad \forall d = 1 \text{ to } ND \quad (8)$$

$$SL_{\min} \leq SL(V^d, P^d, Q^d) < SL_{\max} \quad \forall d = 1 \text{ to } ND. \quad (9)$$

The optimization problem described by (3)–(9) is solved by using the LBIPM [16], as explained in the next section.

IV. SOLUTION STRATEGY

The goal of the proposed solution strategy is to determine the probabilistic behaviour of the problem variables (QS, V, P, Q) via their PDFs. To reach this goal, the LBIPM is used to solve the optimization problem with mean values of load and wind power forecasts. Upon determining the optimal solution, the cumulants of the output variables are obtained using a relationship provided by the inverse of Hessian. Once obtained, they are used to reconstruct the PDF curves by using the Gram-Charlier expansion theory [17].

The main steps in the Cumulant-based optimal capacitor planning approach are:

- 1) Using sensitivity analysis, find the best locations for locating capacitor banks.
- 2) Solve the optimization problem using the LBIPM assuming the mean values of the loads and wind generation forecasts and determine the optimum state.
- 3) Derive the linear relationship between the problem variables (QS, V, P, Q) and the system data (loads and wind generations) using the inverse of Hessian of the Lagrangian function.
- 4) Find the cumulants of the problem variables (QS, V, P, Q) using the linear relationship obtained in Step 3.
- 5) Reconstruct the PDFs of the problem variables (QS, V, P, Q) using the Gram-Charlier expansion theory.

A. Sensitivity Analysis (Step 1)

The solution procedure for minimizing (3) can become computationally difficult and complicated if all variables corresponding to the capacitor (size, location and switching time) need to be determined simultaneously. Strategically, one efficient way to reduce the complexity of the problem is to find the optimal locations (candidate buses) capacitors which lead to the greatest reduction in energy loss prior to solving the optimization problem.

For locating the candidate buses, a sensitivity analysis technique is used to reduce the search space of the optimization problem. The sensitivity analysis pinpoints the buses with maximum effect on the real power losses in the system with respect to the nodal reactive power. Such technique has been widely adapted for capacitor allocation problems. For instance, a novel approach for sensitivity calculations in a radial distribution system was proposed in [18]. In this paper, we follow the idea presented in [18] and adapt it to the power balance described in (4)–(5).

B. LBIPM (Step 2)

The LBIPM is one variant of the interior-point optimization technique that has been commonly used to efficiently solve large power system constrained optimization problems for many years. However, the application of the LBIPM is not only restricted to applications in nonlinear programming problems, but it can also be efficiently applied to Quadratic programming as well as to linear programming problems. In the Capacitor planning problem, the LBIPM provides the inverse of the Hessian matrix of the problem Lagrangian, which can be used to linearly map the probabilistic input variables (load forecast and wind power) into the problem variables (QS, V, P, Q).

To present the features of the LBIPM, consider rewriting the optimization problem presented in (3)–(9) in a simpler form as

$$\text{Minimize : } f(X) \quad (10)$$

$$\text{Subject to: } g(X) = PB \quad (11)$$

$$h(X) \leq 0 \quad (12)$$

where $X = [QS, V, P, Q]$, and $g(X)$ comprises the equality constraints and PB. Here $PB = [PB^1, PB^2, \dots, PB^{ND}]$, where

$$PB^d = \begin{bmatrix} 0 \\ LF^d \cdot PD - PW \\ LF^d \cdot QD \end{bmatrix}. \text{ In the problem represented}$$

above, (10) corresponds to (3), (11) corresponds to (4)–(6), and (12) corresponds to (7)–(9). Writing the Lagrangian function of the above system yields

$$L(U) = f(X) - \lambda^t \times (g(X) - PB) - \mu^t \times h(X) \quad (13)$$

where λ and μ are the Lagrange multipliers, and $U = [X \lambda \mu]^t$ is a vector of primal and dual variables in the optimization problem. Any optimal solution must satisfy the first order Karush-Kuhn-Tucker (KKT) conditions which are written as

$$\nabla_X L(U) = \nabla f(X) - J_g(X)^t \times \lambda - J_h(X)^t \times \mu = 0 \quad (14)$$

$$\nabla_\lambda L(U) = -g(X) + PB = 0 \quad (15)$$

$$\nabla_\mu L(U) = -h(X) = 0 \quad (16)$$

where J_g and J_h are the Jacobian of $g(X)$ and $h(X)$, respectively, and $\nabla f(X)$ is the Gradient of $f(X)$. As the optimization problem (3)–(9) is solved and the system optimum state is estimated, (14)–(16) can be used to obtain the Hessian of the Lagrangian function at the optimum state. This step is presented in the next subsection.

C. Establishing the Linear Relationship Using the Hessian Matrix (Step 3)

The following system can be obtained by applying Newton's method to (14)–(16):

$$\nabla L(U) + H(U) \times \Delta(U) = 0. \quad (17)$$

By rearranging (17) and taking the inverse of Hessian, the vector of changes in the unknown variables can be mapped with the vector of changes in bus generation and loading, i.e., ΔPB , as

$$\Delta U = [-H(U)]^{-1} \times \nabla L(U)$$

$$= [-H(U)]^{-1} \times \begin{bmatrix} \nabla_X L(U) \\ \Delta PB \end{bmatrix}. \quad (18)$$

The Hessian of the Lagrangian is therefore obtained by using the optimum system parameters and, as shown by (17), is linear.

D. Calculating the Cumulants of the Output Variables (Step 4)

The idea proposed in this paper is to determine cumulants of the unknown system variables $X = [QS, V, P, Q]$ from the cumulants of the input variables PB (load demand and wind generations) by developing (18). To incorporate the power system parameters into the above relationship, consider a linear combination of M independent random input variables Z used to create a new random output variable Y as follows [19]:

$$Y = \sum_{m=1}^M C_m \times Z_m \quad (19)$$

where C_m is the m th coefficient in the linear combination. The above expansion can be written in terms of the moment generation function of random variable Y , i.e., $\Phi_Y(s)$, as

$$\begin{aligned} \Phi_Y(S) &= E[e^{s \cdot Y}] = E[e^{s \cdot \sum_{m=1}^M C_m \cdot Z_m}] \\ &= E[e^{s \cdot C_1 \cdot Z_1} \times e^{s \cdot C_2 \cdot Z_2} \times \dots \times e^{s \cdot C_M \cdot Z_M}] \end{aligned} \quad (20)$$

where s is the complex frequency. Assuming that $Z_1, Z_2, Z_3, \dots, Z_M$ are independent, the above relationship can be written as

$$\begin{aligned} \Phi_Y(S) &= E[e^{s \cdot Y}] = E[e^{s \cdot C_1 \cdot Z_1}] \times E[e^{s \cdot C_2 \cdot Z_2}] \\ &\times \dots \times E[e^{s \cdot C_M \cdot Z_M}] \\ &= \Phi_{Z_1}(s \cdot C_1) \times \Phi_{Z_2}(s \cdot C_2) \\ &\times \dots \times \Phi_{Z_M}(s \cdot C_M). \end{aligned} \quad (21)$$

By taking the natural logarithm at both sides of the above equation, the multiplication of the moment generating function of Z_m in the right-hand side of (21) can be converted into the summation of them:

$$\begin{aligned} \ln \{\Phi_Y(s)\} &= \ln \left\{ \Phi_{Z_1}(s \cdot C_1) \times \Phi_{Z_2}(s \cdot C_2) \right. \\ &\quad \times \dots \times \Phi_{Z_M}(s \cdot C_M) \left. \right\} \\ &= \ln \{\Phi_{Z_1}(s \cdot C_1)\} + \ln \{\Phi_{Z_2}(s \cdot C_2)\} \\ &\quad + \dots + \ln \{\Phi_{Z_M}(s \cdot C_M)\}. \end{aligned} \quad (22)$$

Conversely, the cumulants can be used in (18). To do so, the Cumulant generating function $\Psi_Z(s)$ can be written in terms of the moment generating function Φ_Z as [20]

$$\Psi_Y(s) = \ln(\Phi_Z(s)). \quad (23)$$

Expanding (23) gives

$$\Psi_Y(s) = \Psi_{Z_1}(s \cdot C_1) + \Psi_{Z_2}(s \cdot C_2) + \dots + \Psi_{Z_M}(s \cdot C_M). \quad (24)$$

To compute different orders of cumulants, we can set $s = 0$ by taking different order derivatives of the Cumulant generating function. A general equation for the r th-order Cumulant of Y is

$$k_{Y,r} = \Psi_Y^{(r)}(s=0) = C_1^r \cdot \Psi_{Z_1}^{(r)}(0) + C_2^r \cdot \Psi_{Z_2}^{(r)}(0) + \dots + C_M^r \cdot \Psi_{Z_M}^{(r)}(0). \quad (25)$$

Equation (18) provides a linear relation between PB (known) and U (being optimally determined). Using Hessian of (18) to provide this linear relation, one can write determined cumulants of U in terms of cumulants of PB by

$$k_{U,r} = C_1^r \cdot k_{PB1,r} + C_2^r \cdot k_{PB2,r} + \dots + C_n^r \cdot k_{PBn,r} \quad (26)$$

where $k_{U,r}$ is a vector containing the r th cumulants of the system unknown variables (U) and $k_{PBn,r}$ is a vector containing the r th-order cumulants of the random bus generation and loading (PB).

The structure of the Hessian matrix is dependent on the load level modeling. In this paper three different load levels, namely 1-light, 2-medium, and 3-heavy loads, were considered each for determined time duration. That makes the structure of the Hessian matrix consist of three sub-matrices such that the inverse of it contains the coefficients for the linear relationship between system output variables (U) and the input random variables (PB, i.e., loads and wind generations) at each corresponding load level. Similarly, the Cumulant vector that contains the loads and wind generations consists of three sub-vectors as presented follows:

$$\begin{bmatrix} k_{U^{d=1},r} \\ k_{U^{d=2},r} \\ k_{U^{d=3},r} \end{bmatrix} = \left(- \begin{bmatrix} H_{d=1} & & \\ & H_{d=2} & \\ & & H_{d=3} \end{bmatrix}^{-1} \right)^{(r)} \cdot \begin{bmatrix} 0 \\ k_{PB^{d=1},r} \\ k_{PB^{d=2},r} \\ k_{PB^{d=3},r} \end{bmatrix} \quad (27)$$

where the subscripts of the coefficients (H^{-1}) and cumulants (k) can be 1, 2, or 3, standing for Low, Medium, and High, respectively. Consequently, $k_{U^{d=1},r}$, $k_{U^{d=2},r}$, and $k_{U^{d=3},r}$ are the cumulants of the system variables in the light, medium, and heavy load level and $k_{PB^{d=1},r}$, $k_{PB^{d=2},r}$, and $k_{PB^{d=3},r}$ are the cumulants of the system loads and WGs in the prescribed load levels. Once the cumulants of the output variables are computed, their PDFs are reconstructed using Gram-Charlier Expansion theory.

E. Gram-Charlier Expansion Theory (Step 5)

According to the Gram-Charlier Expansion theory, the PDFs of the many distributions can be formed as a series comprised of a known distribution, usually a standard normal distribution, and its derivatives. The proposed solution method makes use of this characteristic to reconstruct the PDFs of the problem variables after their cumulants are defined in terms of the known cumulants. The procedure to obtain the PDFs using the Gram-Charlier Expansion theory is omitted from the paper to ensure conciseness but can be found in [17].

However, it is noteworthy that the order of Hermit Polynomials used in this work is based on the order 3 to 7, which ac-

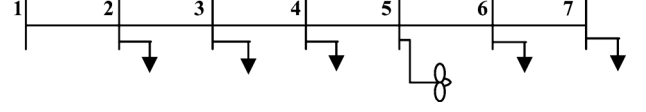


Fig. 3. Single line diagram of the 7-bus distribution test system.

TABLE I
WEIBULL PDF PARAMETERS OF THE WG

Bus No.	Mean Generation (kW)	Shape Factor	Standard Deviation [%]
5	400	2	55% of the Mean

TABLE II
LOAD LEVELS AND DURATION DATA

	Load Level			Capacitor Cost (KC) (\$/kVAr)	Energy Cost (KL) (\$/kWh)
	Low	Medium	High		
Amount (p.u.)	0.5	0.7	1.5	90	0.06
Duration (Hrs)	1000	1000	6760		

cording to our observation and also from [21] gives a better estimation especially for the tail distribution.

V. NUMERICAL RESULTS

The solution method described in the Section IV was applied to a 7-bus and to a 33-bus distribution systems with one and two wind generators connected to them, respectively. In order to assess the accuracy of the solution method, it has been compared with MCS consisting of 5000 samples. The absolute percentage errors of the probabilistic system variables are presented.

A. 7-Bus Test System Case Study

The methodology described in Section IV was first tested on the IEEE 7-bus test system with one wind generator as shown in Fig. 3. The standard deviations of the loads are defined such that the 99% confidence interval is equal to $\pm 10\%$ of the mean loading value.

In this section the results of the Cumulant method are compared with those of 5000 sample point MCS technique. Table I provides the statistical parameters assumed for the WG Weibull distribution. The system and loads data can be found in the Appendix.

The load levels values with their corresponding time duration are provided in Table II.

The sensitivity index calculation is completed to find the best locations for installing the new capacitor banks. The results of the sensitivity analysis yield bus 7 as the best candidate bus for capacitor installation. Table III presents the probabilistic optimal solutions for the real power loss and the cost of energy loss for all durations with and without capacitors comparing the Cumulant method the MCS solutions. Table III also gives the size of capacitors at each candidate bus.

Table III shows a significant decrease in the real power loss and consequently in the annual cost of energy loss after capacitor installation using both methods. The results of the Cumulant method are in close agreement with those of the MCS technique. Table IV gives a comparison of computational time as

TABLE III
OPTIMAL SOLUTIONS AND COMPARISON OF THE TWO METHODS

Method	Time Duration	Capacitors		Real Power Loss (kW)	Total Cost (\$)
		Bus No.	Size (kVAr)		
Base Case Cumulant	TD ₁	0		22.847	4,101,800
	TD ₂			51.588	
	TD ₃			326.09	
Base Case MCS	TD ₁	0		23.708	4,114,500
	TD ₂			52.300	
	TD ₃			326.86	
Cumulant using OPF	TD ₁	7	1738.7	9.9915	2,294,000
	TD ₂			25.068	
	TD ₃			170.48	
MCS using OPF	TD ₁	7	1737.8	10.813	2,300,700
	TD ₂			25.713	
	TD ₃			170.79	

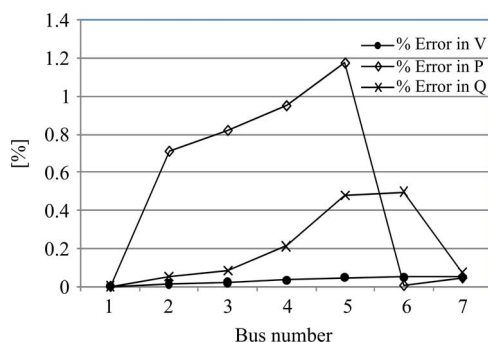


Fig. 4. Error in mean of the output variables using the Cumulant Method and MCSs for TD₃.

well as number of executions between two methods. From this table it can be observed that Cumulant method is significantly faster than MCS since it only needs one run of the capacitor planning problem to provide the optimal solutions as well as the probabilistic information of the all system variables. However, it should be noted that the number of the samples in MCS can be changed based on the required accuracy. The probabilistic optimal results consist of the values for the mean and standard deviation of all system variables, voltage magnitude, active and reactive power flow solutions, and capacitor size in all time durations. Only the results for the High Load (TD₃), which was observed to be the worst case (with highest error), are analyzed in the figures presented in this section. They are shown in Fig. 4 (Mean) and Fig. 5 (Standard Deviation). The mean values for the loads and WGs were taken at the nominal value from the 7-bus system. The results of both mean and standard deviation values are discussed as follows.

- 1) *Mean Values*: From Fig. 4 it can be implied that the percentage errors of the voltage mean values are very small, with a maximum of 0.05%. The percentage errors of the mean value for the other system variables are small and well below 1.2%.
- 2) *Variance Value*: Fig. 5 presents the results of the standard deviation values for all system variables for the Cumulant method and the MCS technique. Fig. 5 shows that the percentage errors of the standard deviation of all variables are

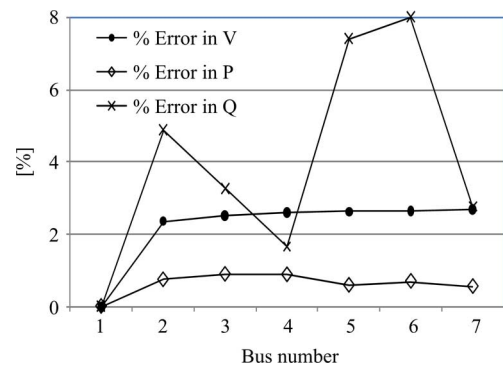


Fig. 5. Error in standard deviation of the output variables using the Cumulant Method and MCSs for TD₃.

TABLE IV
COMPUTATIONAL TIME COMPARISON FOR MCS AND CUMULANT

	Execution Time (Sec.)	Number of executions	Coefficient of Variance [23]
Cumulant	1.134	1	-
MCS	1703.35	5000	0.001478

TABLE V
PARAMETERS OF THE WEIBULL PDF FOR THE WIND GENERATOR

Bus No.	Mean Generation (kW)	Shape Factor	Standard Deviation [%]
2 and 6	500	2	55% of the Mean

below 8%. The maximum percentage error for standard deviation of voltage, active and reactive power and capacitor size are 2.68% at TD₂, 0.90% at TD₂, 7.99% at TD₃, and 0.84% at TD₃, respectively. From this figure it can be said that errors associated with standard deviations of reactive power related quantities are much larger than those associated with the active power. This is due to the fact that the reactive power/voltage magnitude equations are much more nonlinear. Further, connecting capacitor banks to various buses leads to reverse flow of reactive power which causes higher variations in nodal voltage and reactive power in the vicinity buses which results in a higher variance error in comparison to the other variables.

B. 33-Bus Test System Case Study

The second case study is the 33-bus, 32-branch IEEE test system described in [22]. The system, however, is slightly modified to accommodate two wind generators with a mean capacity equal to 500 kW each. The standard deviations of the loads are such that the 99% confidence interval is equal to $\pm 10\%$ of the nominal loading value.

Table V includes the statistical parameters of the Weibull distribution used to represent the WGs. The WGs connected at buses 2 and 6 are identical. The load duration data of the system are those shown in Table II.

The sensitivity analysis procedure outcome produced a recommendation to install four capacitor banks at buses 5, 8, 18, and 33. As in the previous case study, the results obtained using the Cumulant method has been compared with those of 5000

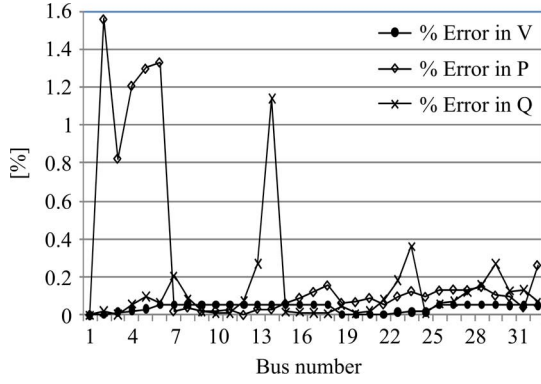


Fig. 6. Error in mean of the output variables using the Cumulant Method and MCS technique for TD₃—33bBus system.

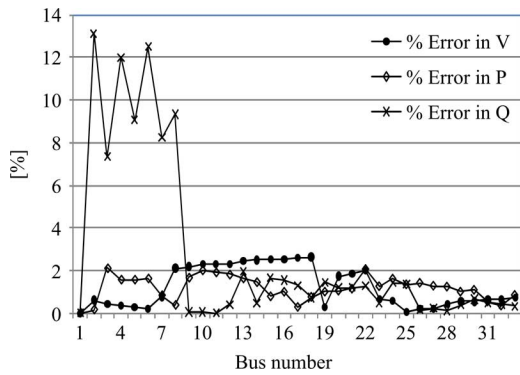


Fig. 7. Error in standard deviation of the output variables using the Cumulant Method and MCS technique for TD₃—33-bus system.

points MCS technique. Table VI presents the probabilistic optimal solutions for the real power loss and the cost of energy loss at each time duration, with and without capacitors. Table V shows that the results of the Cumulant method closely match those of MCS technique. The maximum percentage error for the standard deviation of voltage, active and reactive power and capacitor size are 2.63%, 2.12%, 13.13%, and 5.79%, respectively. Those occur at the third time interval (TD₃). Graphs and tables containing the results for this system are in Figs. 6 and 7.

Table VII presents a comparison of computational time as well as number of executions between two methods. According to this table it can be observed that Cumulant method is significantly faster than MCS. Figs. 7 and 8 show the errors in results of the mean and standard deviation values for all system variables for the Cumulant method compared with the MCS technique for the 33-bus system at TD₃, respectively. The percentage errors of the mean value for the system variables are all below 1.6%, whereas the percentage errors of the standard deviation of all variables are below 14%.

Same as in Fig. 5, in Fig. 7 maximum errors for standard deviation occurs for reactive power related quantities, which is due to the inherent nonlinearity of the problem and these variables.

Figs. 8–10 show that the PDFs have a non-Gaussian behaviours. That can be seen more clearly in Fig. 8. To explain the non-Gaussian behavior of the system random variables at this bus, one may note that one of the WGs is connected at bus 6, and since the WGs were modeled as Weibull distributions,

TABLE VI
OPTIMAL SOLUTIONS AND COMPARISON OF THE TWO METHODS

Method	Time Duration	Capacitors		Real Power Loss (kW)	Total Cost (\$)
		Bus No.	Size (kVAr)		
Base Case Cumulant	TD ₁	0		32.56	5,473,900
	TD ₂			72.11	
	TD ₃			434.38	
Base Case MCS	TD ₁	0		33.17	5,464,700
	TD ₂			72.38	
	TD ₃			433.49	
Cumulant using OPF	TD ₁	8	722.04	18.39	3,601,100
	TD ₂	18	263.30	42.97	
	TD ₃	25	483.27	268.15	
MCS using OPF	TD ₁	8	721.95	18.93	3,583,300
	TD ₂	18	263.15	43.08	
	TD ₃	25	483.30	266.54	

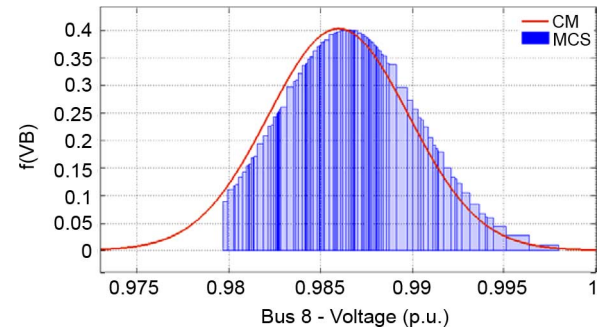


Fig. 8. PDF of voltage at bus 8 reconstructed by Gram-Charlier series.

TABLE VII
COMPUTATIONAL TIME COMPARISON FOR MCS AND CUMULANT

	Execution Time (Sec.)	Number of executions	Coefficient of Variance [23]
Cumulant	4.690	1	-
MCS	3818.19	5000	0.00097

the probabilistic behavior of the systems variables in the buses nearby have a tendency to behave more like a Weibull distribution. The small difference between PDFs of the Cumulant method and the MCS technique in Figs. 8–10 is due to the small error between the standard deviation values obtained from both methods. As it is explained earlier in Section V, the main source of error in standard deviation is the inherent nonlinearity of the problem particularly related to random variables for reactive power/voltage magnitude.

These results clearly imply that the Cumulant method provides results acceptably close to those obtained from MCS. The Cumulant method brings the advantage of computational speed and analytical soundness, as opposed to the exhaustive exploratory nature of MCS technique.

C. 129-Bus Test System Case Study

In order to illustrate the performance of Cumulant method on large systems, it has been tested on a 129-bus test system

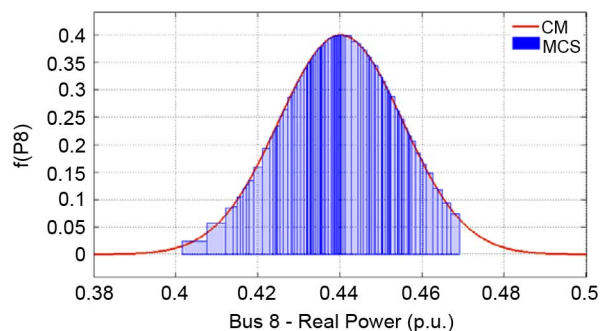


Fig. 9. PDF of real power at bus 8 reconstructed by Gram-Charlier series.

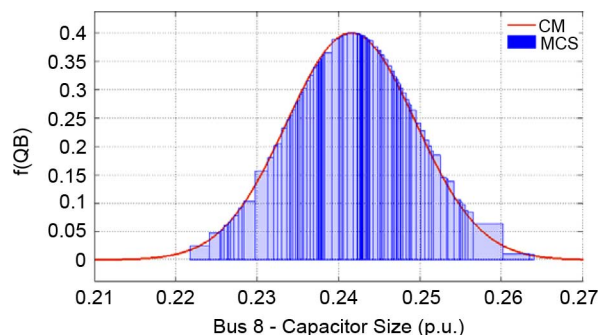


Fig. 10. PDF of capacitor size at bus 8 reconstructed by Gram-Charlier series.

TABLE VIII
AVERAGE ABSOLUTE PERCENTAGE ERROR OF MEAN AND STANDARD
DEVIATION OF THE PROBLEM VARIABLES AT EACH TIME INTERVAL

Variable	Mean			Standard Deviation		
	TD ₁	TD ₂	TD ₃	TD ₁	TD ₂	TD ₃
Voltage	0.085	0.088	0.097	0.693	0.712	10.87
Real Power	12.69	3.649	2.597	0.912	0.911	0.802
Reactive Power	4.214	1.375	4.114	2.052	2.251	19.07
Capacitor Size	0.322	0.245	1.111	2.223	2.077	20.82

TABLE IX
COMPUTATIONAL TIME COMPARISON FOR MCS AND CUMULANT

	Execution Time (Sec.)	Number of executions	Coefficient of Variance [23]
Cumulant	60.8	1	-
MCS	55010.12	5000	0.00652

with nine wind generators. The statistical information of wind generators are same as parameters given in Table V. Table VIII shows the mean absolute percentage error of mean and standard deviation of the problem variables at each time interval.

Table IX presents the computational time comparison for the Cumulant and MCS for the 129-bus system.

VI. CONCLUSION

This paper proposes to use the Cumulant method to relate probabilistic information of load and wind generation in a optimal capacitor planning problem of distribution systems with decision variables. The probabilistic system data are the load forecast and wind availability, which were modeled as Gaussian and Weibull distributions, respectively. Once the best locations for installing the capacitor are determined (via a sensitivity anal-

TABLE X
7 BUS RDS—MEAN VALUE OF THE LOADS

Bus Number	Mean	
	Real Power (kW)	Reactive Power (kW)
1	0	0
2	0.2465	0.197
3	0.2209	0.206
4	0.232	0.286
5	0	0
6	0.701	0.542
7	0.65	0.503

TABLE XI
7 BUS RDS—LINE DATA

From Bus	To Bus	R (Ω)	X (Ω)	Rating (MVA)	System Voltage (kV)
1	2	0.7314	0.7158	100	12.66
2	3	0.8411	0.8231	100	12.66
3	4	0.8411	0.8231	100	12.66
4	5	1.0605	1.0379	100	12.66
5	6	1.197	0.808	100	12.66
6	7	0.8542	0.5766	100	12.66

ysis), the optimization problem is formulated and solved using the LBIPM. This solution provides the Hessian matrix of the Lagrangian, which is inverted to provide a linear mapping between the cumulants of bus powers (loads and WGs) and decision variables (capacitor values, voltage, reactive and tap settings). Once the cumulants of all decision variables are obtained, their PDF scan be constructed using the Gram-Charlier expansion. The method was implemented and tested on a set of 7-bus, 33-bus, and 129-bus systems. In order to illustrate the efficiency and accuracy of the Cumulant method, the results were compared with those of MCS technique with 5000 samples. The errors were found to be acceptable. The proposed probabilistic algorithm has much less computational burden and complexity, making it very practical and advantageous. Moreover, the PDFs provided by the proposed solution offers great insight to power system planners about influence of WG uncertainty on the optimal solution.

While the results of this paper are convincingly promising, further research and development is required to overcome the limitations of using approximation of Gram Charlier Expansion theory. As one of these limitations, one may recognize that if discrete random variables are considered to model possible related uncertainties, the Gram-Charlier Expansion theory may become unreliable. Moreover, for using the proposed method in a different probabilistic problem, in order to identify the best order of Hermit Polynomials which gives the best PDF approximations, the final result needs to be compared with those obtained from MCS in order to certify accuracy.

Efforts must also be devoted to contain any potential increase in errors that could creep in due to the inherent nonlinear nature of the problem.

APPENDIX

Table X presents the mean values used for the load demands, and Table XI presents the feeder data.

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